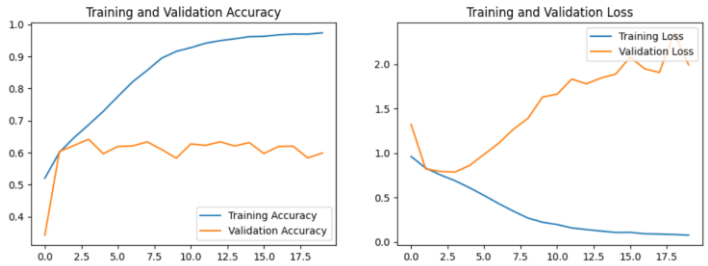
Model 1: (EPOCHS = 20, LR = 0.001)

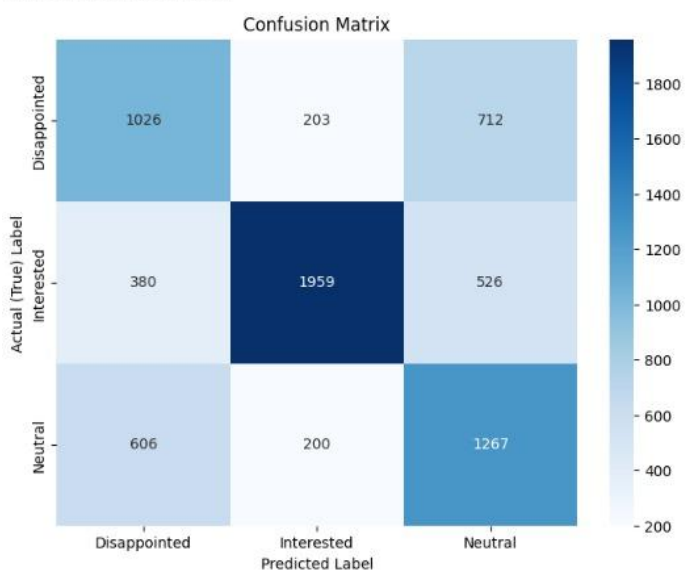
```
Evaluating model on validation data....
108/108 ━━━━━━━━━━━━━━━━━━━━ 4s 36ms/step - accuracy: 0.5988 - loss: 1.9943
Final Validation Accuracy: 59.88%
```




```
Gathering predictions from validation set...
Predictions gathered...
Classification Report
------------------------------
```

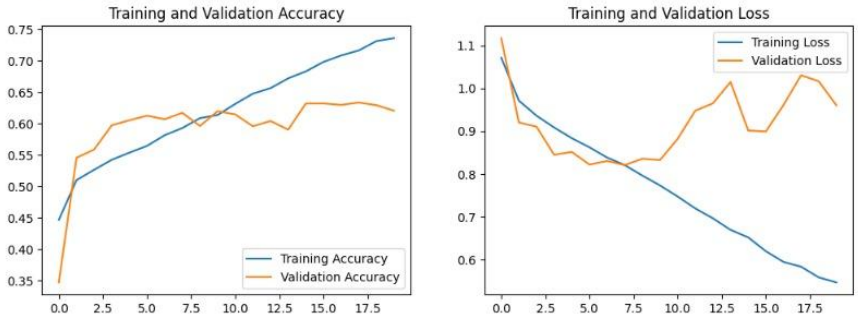
	precision	recall	f1-score	support
Disappointed	0.51	0.53	0.52	1941
Interested	0.83	0.68	0.75	2865
Neutral	0.51	0.61	0.55	2073
accuracy			0.62	6879
macro avg	0.62	0.61	0.61	6879
weighted avg	0.64	0.62	0.63	6879

```
Generating Confusion Matrix...
------------------------------
```



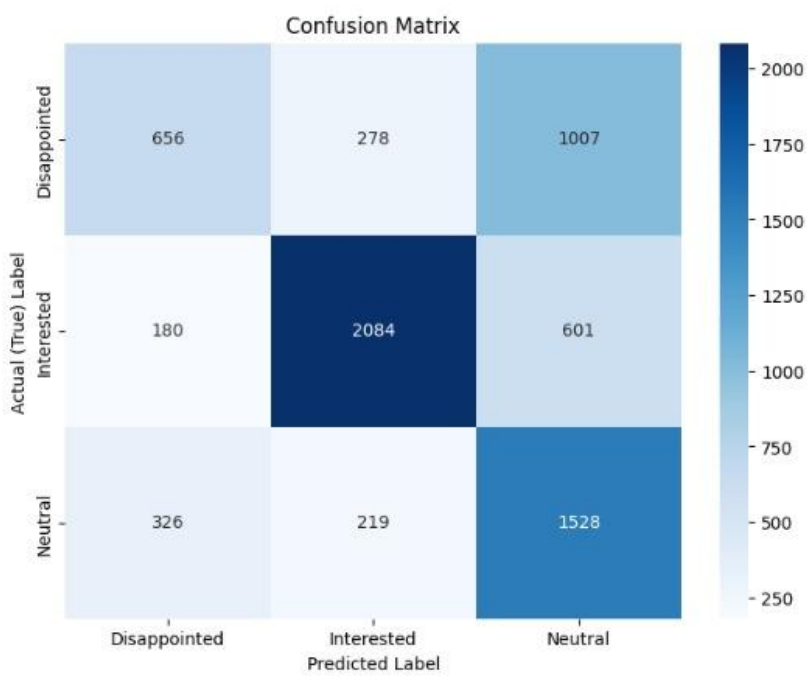
Model 2: (EPOCHS = 20, LR = 0.001, DROPOUT = 0.5)

```
Evaluating model on validation data....
108/108 ━━━━━━━━━━━━━━━━━━━━ 2s 21ms/step - accuracy: 0.6204 - loss: 0.9605
Final Validation Accuracy: 62.04%
```



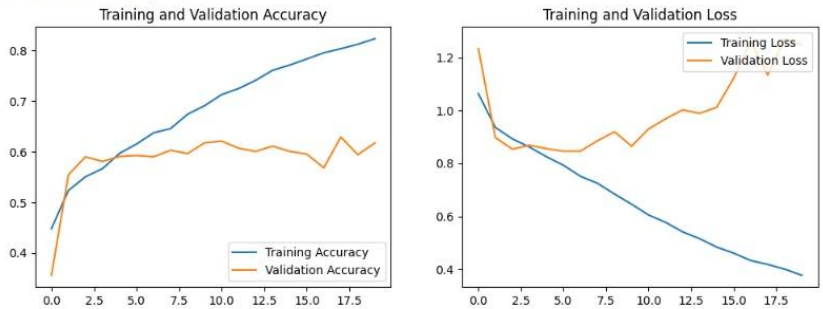

```
Gathering predictions from validation set...
Predictions gathered...
Classification Report
------------------------------
```

	precision	recall	f1-score	support
Disappointed	0.56	0.34	0.42	1941
Interested	0.81	0.73	0.77	2865
Neutral	0.49	0.74	0.59	2073
accuracy			0.62	6879
macro avg	0.62	0.60	0.59	6879
weighted avg	0.64	0.62	0.61	6879



Model 3: (EPOCHS = 20, LR = 0.001, IMG_SIZE = 40)

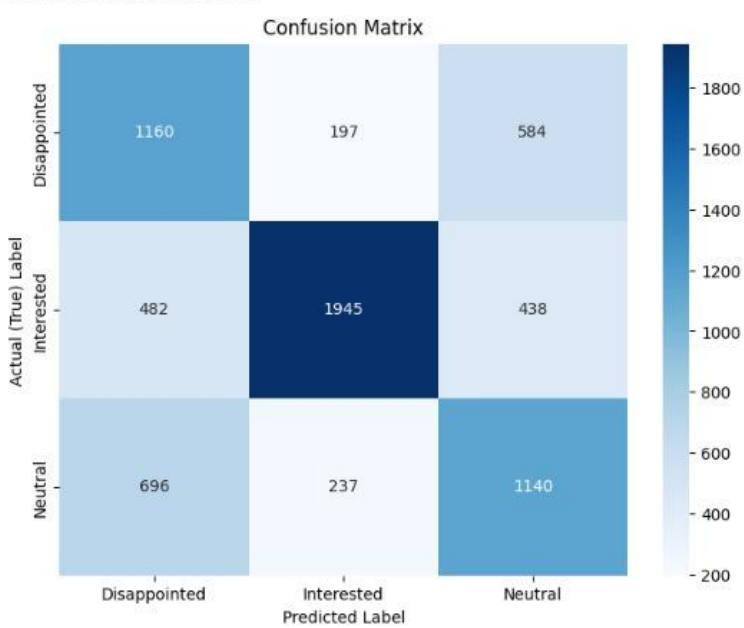
```
Evaluating model on validation data....
108/108 ━━━━━━━━━━━━━━━━━━━━ 2s 15ms/step - accuracy: 0.6171 - loss: 1.2467
Final Validation Accuracy: 61.71%
```




```
Gathering predictions from validation set...
Predictions gathered...
Classification Report
------------------------------
```

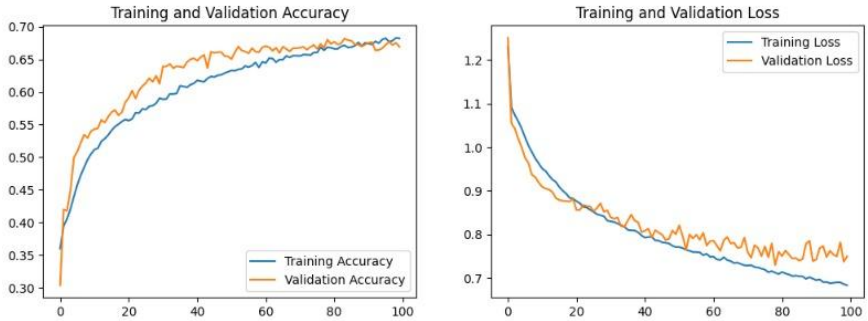
	precision	recall	f1-score	support
Disappointed	0.50	0.60	0.54	1941
Interested	0.82	0.68	0.74	2865
Neutral	0.53	0.55	0.54	2073
accuracy			0.62	6879
macro avg	0.61	0.61	0.61	6879
weighted avg	0.64	0.62	0.62	6879

```
Generating Confusion Matrix...
------------------------------
```



Model 4: (EPOCHS = 100, LR = 0.0001, DROPOUT = 0.5) Augmentation

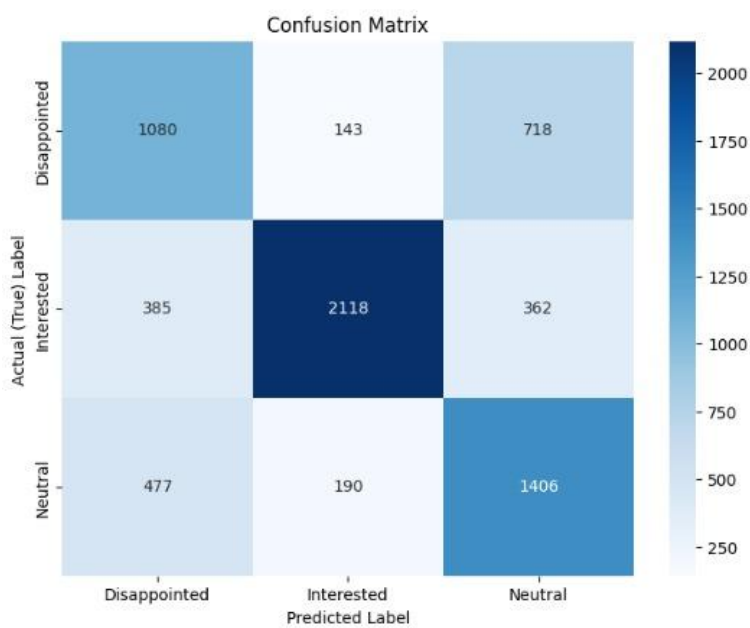
```
Evaluating model on validation data....
108/108 ━━━━━━━━━━━━━━━━━━━━ 3s 26ms/step - accuracy: 0.6693 - loss: 0.7495
Final Validation Accuracy: 66.93%
```




```
Gathering predictions from validation set...
Predictions gathered...
Classification Report
------------------------------
```

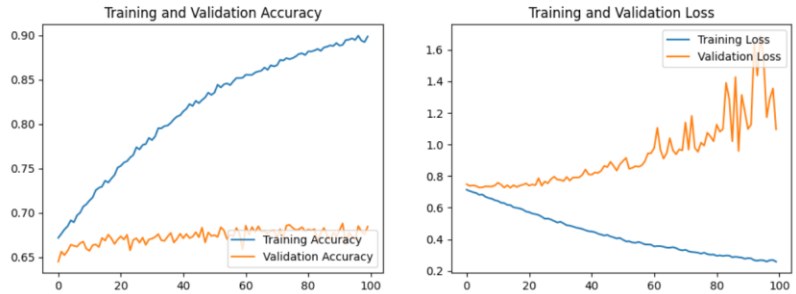
	precision	recall	f1-score	support
Disappointed	0.56	0.56	0.56	1941
Interested	0.86	0.74	0.80	2865
Neutral	0.57	0.68	0.62	2073
accuracy			0.67	6879
macro avg	0.66	0.66	0.66	6879
weighted avg	0.69	0.67	0.67	6879

```
Generating Confusion Matrix...
------------------------------
```



Model 5: KerasTuner Best (100 Epochs, LR=0.0001, Data Augmentation)

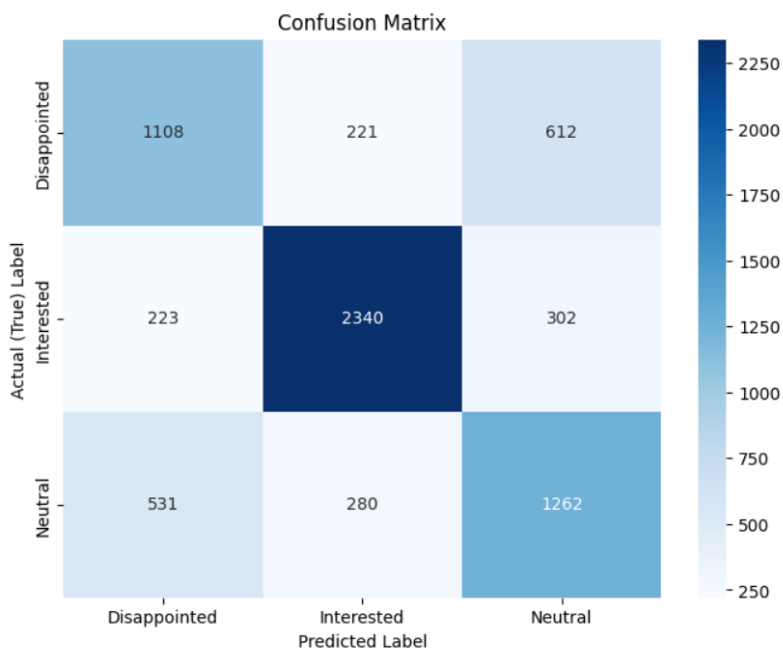
```
Evaluating model on validation data....
108/108 ━━━━━━━━━━━━━━━━━━━━ 3s 32ms/step - accuracy: 0.6847 - loss: 1.0960
Final Validation Accuracy: 68.47%
```




```
Gathering predictions from validation set...
Predictions gathered...
Classification Report
------------------------------
```

	precision	recall	f1-score	support
Disappointed	0.60	0.57	0.58	1941
Interested	0.82	0.82	0.82	2865
Neutral	0.58	0.61	0.59	2073
accuracy			0.68	6879
macro avg	0.67	0.67	0.67	6879
weighted avg	0.69	0.68	0.69	6879

```
Generating Confusion Matrix...
------------------------------
```



Comparison Table:

Model Variety	Validation Accuracy	F1-Score (Marco avg)	Precision	Confusion Matrix Observation
Model 1: (EPOCHS = 20, LR = 0.001)	59.88%	0.61	0.62	The validation loss graph shows it rising sharply. Strong confusion between "Disappointed" and "Neutral".
Model 2: (EPOCHS = 20, LR = 0.001, DROPOUT = 0.5)	62.04%	0.59	0.62	"Disappointed" recall dropped to 0.34. Massive confusion, with 1007 "Disappointed" images predicted as "Neutral".
Model 3: (EPOCHS = 20, LR = 0.001, IMG_SIZE = 40)	61.71	0.61	0.61	No improvement. Performed almost identically to Model 1. Still high confusion between "Neutral" (696 errors) and "Disappointed" (584 errors).
Model 4: (EPOCHS = 100, LR = 0.0001, DROPOUT = 0.5) Augmentation	66.93%	0.66	0.66	Good, stable model. The loss/accuracy plots show no overfitting. This is the second most balanced model, with the best F1-Score.
Model 5: KerasTuner Best (100 Epochs, LR=0.0001, Data Augmentation)	68.47%	0.67	0.67	Highest accuracy, but overfit. The loss plot shows clear overfitting after epoch 80. This is the most balanced model, with the best F1-Score.

Conclusion:

The most significant improvements came from a combination of data augmentation, a slower learning rate (0.0001), and longer training (100 epochs).

Model 4 was the most stable and balanced model. It achieved a 66.93% accuracy and the highest macro F1-Score (0.66). Critically, its training graphs show a perfect, stable fit with no overfitting.

Model 5, the KerasTuner result, achieved the highest overall accuracy at 68.47%. It did this by becoming an expert at the "Interested" class (85% recall). However, this came at a cost: its training graphs clearly show it began to overfit after approximately 80 epochs, with validation loss rising sharply. This overfitting made it perform worse than Model 4 on the other two classes.

Key Limitation:

The primary challenge across all models is the strong confusion between the "Disappointed" and "Neutral" classes. For example, in the best-performing Model 5, it still misclassified 476 "Neutral" images as "Disappointed" and 678 "Disappointed" images as "Neutral".